

What Did Ten Points of Mean Recall Buy? Separating Group-Level from Case-Level Gains in Scene Graph Generation

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Two frozen models, one cell: relabelling within ϕ removes case identity and nothing else

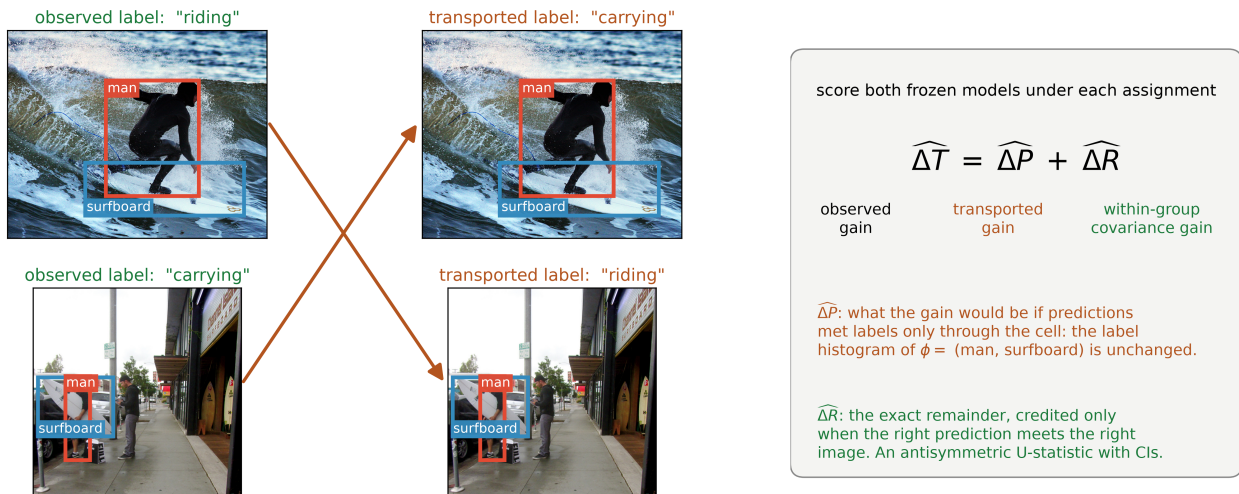


Figure 1. **Within-group relabelling.** Two Visual Genome relations share the subject–object pair (man, surfboard). Swapping their labels keeps the pair’s predicate histogram exactly as it was but breaks the link between each image and its own label. Scoring two frozen models under the observed and the swapped labels splits their score gain exactly: $\widehat{\Delta P}$ is the part that survives the swap — earned at the level of the group — and $\widehat{\Delta R}$ is the part that needs the right label on the right image.

Abstract

001 Debiasing methods for scene graph generation are judged
002 by mean recall, and one of the best known, Total Direct Ef-
003 fect (TDE), raises it on Visual Genome from 14.6 to 24.8.
004 What did those ten points buy? A model raises mean re-
005 call either by predicting rare predicates more often within
006 each subject–object pair, or by recognising which individ-
007 ual images actually show them, and the metric credits both
008 alike. We give an exact way to tell them apart. Scoring
009 each prediction against the label of a different relation from
010 the same subject–object group keeps the group’s predicate
011 frequencies but breaks the link between each relation and
012 its own label; this splits the difference between two frozen
013 models, in a proper score or in mean recall itself, into a

group-level part and a case-level covariance, with no fitted
baseline, in $O(NK)$ time, with image-clustered intervals.
On TDE’s released checkpoint, 87% of the mean-recall gain
is group-level and almost none of it comes from the fif-
teen rarest predicates. A one-line logit adjustment of the
baseline, which leaves the ranking of relations within ev-
ery group unchanged, recovers nine of the ten points with
the same split, at under a tenth of TDE’s cost in recall,
and a threshold-free within-group AUC finds no discrimina-
tion that TDE gained. A second released method, IETrans,
gains 21 points, 97% group-level, while its case-level part
falls under both proper scores. The converse also occurs:
a CLIP model reading image crops, ranked last on accu-
racy and on the proper score, carries more case-level co-
variance than the geometry model ranked above it, though

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029 *a threshold-free rank test cannot separate the two.*

030 1. Introduction

031 Scene graph generation is evaluated on a benchmark whose
032 labels are unusually predictable from something every
033 model sees. Given the ordered object pair (man, surf-
034 board), the predicate on Visual Genome [17] is very of-
035 ten riding, and a lookup table of training frequencies is fa-
036 mously competitive [42]. Recall@ K rewards models that
037 echo this prior, so the field now leads with mean recall,
038 which averages recall over predicates and so rewards getting
039 the rare ones [4, 35]. Debiasing methods are now judged by
040 how far they raise it. Total Direct Effect (TDE) [36], among
041 the most cited of them, raises mean recall@50 of a MOTIFS
042 model from 14.6 to 24.8 on the canonical VG150 split, in
043 our evaluation of the released checkpoint.

044 What did those ten points buy? There are two ways to
045 raise mean recall, and the metric cannot tell them apart. A
046 model can move probability towards rare predicates *within*
047 *each subject-object pair* — say carrying more often for
048 (man, surfboard) — without becoming any better at de-
049 ciding *which* images show a man carrying rather than rid-
050 ing. Or it can learn to make that decision: put more proba-
051 bility on the correct predicate for the individual image, be-
052 yond what the object pair implies. The first is a change in
053 the model’s group-level predictions; only the second is bet-
054 ter recognition. Every existing remedy — head/body/tail-
055 stratified recall [20], independent mean recall [22], zero-
056 shot recall [24], rarity-stratified accuracy [16] — slices one
057 model’s performance differently. None attributes the gain
058 *between* two models to one route or the other.

059 **An exact split.** We separate the two routes with one op-
060 eration (Fig. 1). Take two frozen models and a test set
061 grouped by a discrete feature both models observe, here the
062 subject-object pair. Score each model’s prediction for one
063 relation against the label of a *different* relation in the same
064 group. This relabelling keeps each group’s predicate fre-
065 quencies, so gain earned at the level of the group survives
066 it, and breaks the link between an image and its own label,
067 so gain earned on individual cases does not. The observed
068 gain therefore splits exactly into a *group-level* part $\widehat{\Delta P}$ and
069 a *case-level* remainder $\widehat{\Delta R}$, which is a within-group covari-
070 ance between the change in the models’ predictions and the
071 label (Theorem 1). The split holds in the observed sam-
072 ple for any proper score and for mean recall itself, needs
073 no fitted reference model, held-out fold or tuning param-
074 eter, costs $O(NK)$ on cached predictions, and comes with
075 image-clustered intervals.

What it finds. We re-ran TDE’s own released checkpoint
with the authors’ code and replayed the official evaluator
exactly, so that recall can be computed under any labelling,
and split mean recall itself (Sec. 5). Of the ten points, 87%
is group-level, the body of the predicate distribution gains,
the head loses, and the fifteen rarest predicates contribute
almost nothing: TDE recalls one of them, at 0.016. Re-
call of parked on rises from 0 to 0.886 because TDE now
says parked on for (car, street). The case-level remainder
needs care, and handling it is part of the method: a thresh-
olded metric earns case-level credit whenever its operating
point moves. Subtracting the training predicate prior from
the baseline’s logits [26], which leaves every within-group
ranking exactly as it was, recovers nine of the ten points
with the same split, at under a tenth of TDE’s cost in $R@50$;
against it, TDE’s case-level part is -0.0008 with an inter-
val straddling zero, and a threshold-free within-group AUC
finds no discrimination TDE gained. The split also sepa-
rates methods: IETrans [43], which relabels training data
within object pairs, gains 21 points of mean recall, 97%
group-level, and loses case-level score under both proper
scores. The converse appears too. A model reading image
crops through a frozen CLIP encoder [29], which every ag-
gregate measure ranks last, gains $+0.00464$ of case-level
covariance over the geometry-based model ranked above
it, with calibration matched and an interval excluding zero
(Sec. 6): its deficit is entirely group-level. That gain does
not make it the better ranker, though, and the within-group
AUC does not separate the two models; the disagreement is
itself part of what Sec. 6 reports.

Contributions.

- Audits of two published debiasing methods’ released checkpoints that split their mean-recall gains exactly: TDE’s ten points 87% group-level, with a case-level part no larger than a ranking-preserving logit adjustment’s and no gain in within-group discrimination, and the mechanism behind it; IETrans’s 21 points 97% group-level, with a case-level loss — and a reversal in which the model ranked last carries the most case-level covariance among relations sharing an object pair.
- An exact, model-free decomposition of the difference between two frozen models into group-level and case-level parts, for every Bregman score and for label-weighted hit rates such as mean recall, computed in one pass over cached predictions, with cluster-robust inference, and paired with an operating-point control and a threshold-free AUC.
- Validation that the case-level part is exactly zero when it should be, covers at the nominal rate under image clustering, and separates calibration, prior and shortcut effects, with a protocol for the one confound it does not separate.

What is not claimed. Every scene-graph result is Pred-Cls, where object boxes and classes are given and every model sees the cell; in SGCLs and SGDet the cell is itself predicted, and the split as stated does not apply. The split is conditional on the declared grouping: $\widehat{\Delta R}$ measures case-level discrimination *among relations sharing an object pair*, and credits any signal varying inside a pair, including a shortcut nobody recorded. It is not invariant to recalibration — rescaling a model’s confidence moves score between the two parts while changing no ranking — which is why the comparisons whose models differ visibly in confidence, TDE’s and the CLIP model’s, are read confidence-matched. For mean recall the split is of the micro-averaged version, which pools relations across images; it tracks the official per-image average but is not it. And the proper-score split’s headline holds at the configuration we argue for, not at every configuration; Appendix O reports all of them. These limits are taken up in Sec. 7.

2. Related work

Bias and its evaluation in scene graph generation. Scene graph generation was posed as structured prediction over a whole image, solved by message passing between object and relation nodes [39]; MOTIFS then showed a frequency predictor of the predicate given the object pair to be surprisingly competitive [42]. Mean recall became the headline metric, and debiasing methods now span the causal TDE intervention [36], loss re-weighting and group-wise learning [9, 11], predicate-rebalancing message passing [20], relabelling of the training data [43] and, outside scene graphs, post-hoc logit adjustment [26]; the line continues [18, 25], and the metrics themselves have been shown to reward pathologies [22], a diagnosis that recurs in open-vocabulary settings [21]. Benchmark audits outside scene graphs find the same pattern: VQA models collapse when answer priors change between train and test [1], GQA-OOD stratifies accuracy by answer rarity within question groups [16], and multimodal models exploit non-visual shortcuts [12, 44]. Gains can also be credited to a mechanism that did not produce them: grounding-based VQA methods keep their out-of-distribution improvement when the grounding signal is irrelevant [33], and methods can exploit how an out-of-distribution split was built [37]. Long-tailed recognition faces the same question for re-weighting and logit adjustment [3, 26]. All of these characterise a *single* model. We attribute the gain *between* two.

Proper-score decompositions. Conditional on a grouping, a proper score splits into reliability and resolution terms [7, 27], for general proper scores [2]. Our case-level term is not that resolution term: it descends from the *covariance* decomposition of the Brier score due to Yates [40],

which a monotone recalibration moves while the classical resolution is invariant to it. The two coincide only when both models are calibrated within every group. Variance estimators exist for single-model decompositions [34], but not for a contrast between two models scored on the same data. The grouping loss [28] is the single-model relative of our case-level part: what calibration leaves behind because inputs sharing a confidence level differ; ours contrasts two models within groups declared in advance.

Comparing two predictors. Inference on a paired score difference is the Diebold–Mariano test [10] and its conditional form [13]; at a single group our total-gain interval is a clustered Diebold–Mariano test (Corollary 3), so the decomposition adds a split to what those tests already report. DeLong *et al.* [8] compare correlated ROC areas with a paired U-statistic, the same device we use for a group-conditional score gain. The case-level term is an order-two U-statistic [15], with a Hájek projection supplying its influence function [32].

3. Splitting a score gain

Setup. Let $q_0(\cdot | x)$ and $q_1(\cdot | x)$ be the predictive distributions of an old and a new frozen model over K labels, and $S(q, y)$ a bounded proper score, larger being better. Our default is the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, negative Brier loss up to a label-only constant; the log score is the second case we report. Let $C = \phi(X)$ be a discrete grouping both models observe, declared before the audit — for relation prediction, the ordered subject–object class pair. We call a value of C a *cell*. The gain vector

$$H_x(y) = S(q_1(\cdot | x), y) - S(q_0(\cdot | x), y), \quad y = 1, \dots, K, \quad (1)$$

can be evaluated at a counterfactual label without rerunning either model, and that is all the construction needs.

Transport. For a cell c , let $\Delta T_c = \mathbb{E}[H_X(Y) | C = c]$ be the observed gain and $\Delta P_c = \mathbb{E}_{H|c} \mathbb{E}_{Y'|c}[H_X(Y')]$ the gain when each case is scored against an *independent* label Y' drawn from the same cell, and set $\Delta R_c = \Delta T_c - \Delta P_c$. Transport keeps the cell and its label distribution and removes only the pairing between a case and its own label.

Theorem 1 (Transport and covariance identities). *For any integrable score contrast H ,*

$$\Delta T_c = \Delta P_c + \Delta R_c, \quad (2)$$

$$\Delta R_c = \sum_{y=1}^K \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | C = c). \quad (2)$$

For the quadratic score, with $\Delta q_y(X) = q_1(y | X) - q_0(y | X)$,

$$\Delta R_c = 2 \sum_{y=1}^K \text{Cov}(\Delta q_y(X), \mathbb{1}\{Y = y\} | C = c). \quad (3)$$

Consequently, if the gain vector is a function of c alone, $\Delta R_c = 0$.

Eq. (3) is what the split means. The new model earns case-level credit only insofar as its probability *change* rises on the label the individual case actually carries, relative to other cases in the same cell. Anything constant within a cell — shifting probability towards rare predicates for every (man, surfboard) relation alike — lands in ΔP , which is why we call it the group-level part. It is *not* simply “prior fit”: under the quadratic score ΔP_c is the improvement in how close the model’s mean forecast sits to the cell’s label frequencies *minus* the increase in the dispersion of its forecasts (Proposition 1). The covariance form is due to Yates [40] and generalises: for every Bregman score, including the log score, ΔR_c is the same covariance with Δq_y replaced by the contrast of the y -th gradient coordinate (Theorem 2). All proofs are in the supplementary.

Why it matters which part a gain is in. In one cell with three cases labelled 1 and one labelled 2, against an old model predicting $(\frac{1}{2}, \frac{1}{2})$, a model predicting the cell’s frequencies $(\frac{3}{4}, \frac{1}{4})$ for every case gains $+\frac{1}{8}$ of quadratic score, all group-level; a model equally far from the old one but pointed at each case’s own label gains $+\frac{3}{8}$, with a case-level part of $+\frac{1}{2}$. Shift the cell’s labels to $(\frac{1}{4}, \frac{3}{4})$ and the first falls to $-\frac{3}{8}$, worse than the model it replaced, while the second stays at $+\frac{3}{8}$. In general, with $m_y(z) = \mathbb{E}[H_X(z) | Y = y, C = c]$, $\Delta R_c = \sum_{y < z} p(y)p(z) D(y, z)$ where $D(y, z) = m_y(y) - m_y(z) + m_z(z) - m_z(y)$: a label shift fixes every D and moves only the weights, so a case-level gain is less exposed to a shift than a group-level one, though with three or more labels it can still reverse (Appendix C).

Estimator. In a cell with $n_c \geq 2$ cases, write $H_i(y) = H_{x_i}(y)$ and estimate

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_i H_i(y_i), \quad \widehat{\Delta P}_c = \frac{1}{n_c(n_c-1)} \sum_{i \neq j} H_i(y_j), \quad (4)$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j} A_{ij}, \quad (5)$$

$$A_{ij} = \frac{1}{2} \{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \quad (6)$$

These satisfy $\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c$ exactly in every sample, and $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic [15] for ΔR_c (Theorem 3). Transporting a case only to the *other* cases’ labels matters: fitting the cell’s label frequencies on

all n_c cases, the natural plug-in, shrinks every cell’s case-level gain by exactly $(n_c - 1)/n_c$, halving it at $n_c = 2$ (Proposition 4). With m_{cy} the count of label y in cell c , case i ’s transported score is

$$P_i = \frac{\sum_y m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}, \quad (7)$$

so the whole estimator costs $O(NK)$ on cached predictions, with no fitted model, held-out fold or smoothing parameter. Cells are weighted by size; singletons cannot identify a within-cell counterfactual and are excluded and reported.

Inference. The Hájek influence contributions of the U-statistic (Appendix F), summed within each image, give one-way cluster-robust intervals, since relations from one photograph are not independent; the estimator is asymptotically normal under bounded scores and non-dominating clusters (Theorem 4), and at a single cell the total-gain interval is a clustered Diebold–Mariano test [10] (Corollary 3).

Confidence matching. One property of the split must be part of the procedure rather than a caveat. Because ΔR is a covariance between probability *movements* and labels, rescaling a fixed model’s confidence moves score between the two parts while changing no ranking, no top-1 decision and no information about any case (Sec. 4 measures how much). Two released checkpoints that differ in calibration therefore cannot be compared on their raw outputs. Whenever compared models differ visibly in confidence we split the images at random, fit each model a single temperature by maximum likelihood on one half, and run the split on the other half with each model’s own temperature applied. The protocol uses no label from the audited half.

Splitting mean recall. Nothing in Theorem 1 or the estimator uses properness: both hold for any contrast $H_x(y)$ that can be evaluated at a counterfactual label. Mean recall@ k is one. Let $h_x^{(m)}(y) \in \{0, 1\}$ say whether model m would recall relation x among its top k if its label were y ; replaying the benchmark’s matching rules on cached scores gives it for every y (Appendix O). Pooling relations, mean recall is $\sum_i w(y_i) h_{x_i}(y_i)$ with $w(y) = 1/(Kn_y)$, so $H_x(y) = Nw(y)[h_x^{(1)}(y) - h_x^{(0)}(y)]$ makes $\widehat{\Delta T}$ its change; transport preserves every n_y , and restricting w to one tier or predicate splits that share exactly. One difference matters for reading the result. A hit is a thresholded decision, so moving a model’s operating point — subtracting a log-prior from its logits, say — flips first the cases it already scored as most like the favoured label, and those carry that label more often: a case-level part with no new discrimination. We therefore read the split against an *operating-point control*, the old model’s logits shifted by the training log-prior [26],

and pair it with a threshold-free test, the *within-cell AUC*: the probability that a case labelled y has larger log-odds $\log q(y) - \log q(z)$ than one labelled z in the same cell, averaged with weight $n_c(y)n_c(z)$ as in the pairwise form above. A temperature, and any change constant within a cell, leaves it unchanged.

Choosing the grouping. The split is conditional on ϕ . Merging cells changes it by an explicit between-cell covariance term of either sign (Proposition 2), so the finest grouping that the benchmark’s construction defends is the conservative choice. A relevant variable left out of ϕ is the same term with the fully adjusted cell as the fine partition, and Proposition 3 bounds it by a single elicited correlation, in the spirit of the robustness value of Cinelli and Hazlett [5].

4. Does the split measure what it claims?

A released checkpoint cannot validate the split, because what the model can see is unknown. We validate it where the answer is known, first in simulation and then on Visual Genome predictors whose information content we control. All intervals are 95%.

Simulation. With 24 cells over five labels, an old model equal to the cell prior and a new one interpolating towards an oracle, the estimated case-level gain grows linearly with the injected signal; under a within-cell null the randomization test rejects in 0.030 of 200 datasets at level 0.05 (mean estimate -8.97×10^{-5}). For the interval we use on real data, we simulate relations clustered into images with a shared per-image effect and Zipf-distributed cell sizes, most cells holding two cases: the image-clustered interval covers in 94.0% of 500 runs, against 88.6% for an interval that treats relations as independent.

Three designs check that the two parts separate what they claim to. A *prior-only* improvement, where the new model corrects a noisy estimate of the cell prior, gives a case-level gain centred at zero (0.0002, sd 0.0035). An *unrecorded within-cell shortcut* — a binary feature varying inside each cell that the new model uses — is credited to the case-level part (+0.041, sd 0.004): the split measures all within-cell signal relative to the declared grouping, shortcut or not. And a *calibration-only* change, where the new model is a temperature-softened copy of an overconfident one carrying identical case-level information, is credited a large spurious case-level *loss* of -0.488 on raw outputs, which the confidence-matching protocol of Sec. 3 reduces to -0.0003 (sd 0.0017), which is why that protocol is part of the procedure.

Controlled predictors on VG150. We train predicate classifiers whose inputs we control on a VG150-style split

we reconstruct from the released Visual Genome annotations, with the same 150-object and 50-predicate vocabulary and images divided 70/30 between fitting and audit (Appendix N.1). It is not the canonical split of Sec. 5, which is why its test set is larger; models trained here from annotations need not reproduce a released split. The predictors are: a frequency table that sees only the class pair (FREQ), an MLP on class embeddings (MLP-CLASS), and MLPs adding twelve box-geometry features (MLP-SPATIAL, MLP-SPATIAL+). The audit covers 227,337 of 229,605 test relations (99.0%) in 6,346 class-pair cells.

MLP-CLASS beats FREQ by 0.03386 of quadratic score and the split assigns *all* of it to the group-level part: the case-level gain is 0.00000 with a zero-width interval and randomization $p = .724$. That zero is algebraic, not empirical — both models are functions of the class pair alone, so Theorem 1 forces it — and it is the sharpest check that nothing is credited to case-level evidence when none exists. Adding box geometry then yields a case-level gain of 0.01439 (CI [0.01373, 0.01504], $p = .002$) that no aggregate score isolates. Against MLP-CLASS the total gain of 0.00885 even *understates* it, because the group-level part deteriorates by 0.00554: geometry moved the model’s probability towards each relation’s own label within its pair, and made it a slightly worse fit to the cell frequencies at once.

5. What TDE’s ten points bought

Tang *et al.* [36] release a causal MOTIFS PredCls checkpoint that yields two models from one set of weights: the baseline (MOTIFS-SUM) and its Total Direct Effect counterpart (MOTIFS-TDE). The baseline’s predicate logit is a sum of three terms: a visual term from the union region, a context term, and a frequency prior indexed by the object pair, which in PredCls is fixed by the subject–object cell. TDE subtracts the same sum evaluated with an averaged context. The visual and frequency terms enter both halves unchanged and cancel, so TDE’s output is the context term minus its averaged counterpart: it removes the frequency prior, constant within a cell, and also the image-specific visual term, which is not. We ran the released checkpoint once with the authors’ code on the canonical VG150 test split, saving every branch’s logits for every object pair, and re-scored them offline with an exact replay of the official evaluator’s matching (Appendix O). The replay reproduces the official numbers to the fourth decimal: the baseline reaches $R@50 = 0.6612$ with $mR@50 = 0.1459$, TDE reaches $mR@50 = 0.2476$ while $R@50$ falls to 0.4588, and without the graph constraint TDE’s mean recall *falls*, from 0.3260 to 0.2981. The audit covers 183,642 relations over 26,446 images in 7,127 subject–object cells; 98.9% of the relations fall in the 5,044 cells holding at least two.

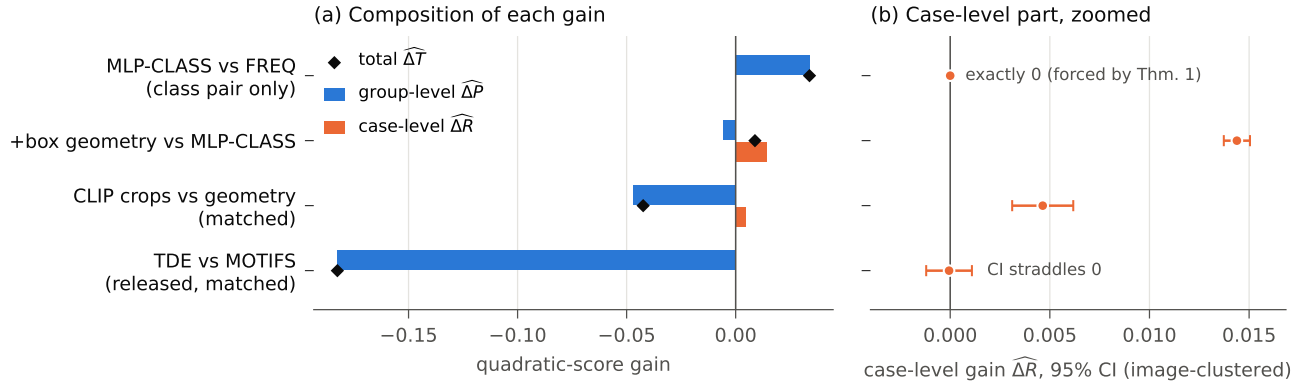


Figure 2. **Where each proper-score gain lives** (quadratic score). (a) Total gain $\widehat{\Delta T}$ (diamond) split exactly into group-level $\widehat{\Delta P}$ and case-level $\widehat{\Delta R}$. (b) $\widehat{\Delta R}$ with its image-clustered 95% interval. Class-pair-only models differ purely at group level (Theorem 1); geometry and CLIP crops buy case-level gain, CLIP despite losing overall; TDE’s change is group-level. CLIP and TDE rows are confidence-matched.

Table 1. **Mean recall@50 split** (graph constraint, micro-averaged over identified relations; official $mR@50$ in the first column). Each row is a model against the MOTIFS baseline, except TDE–LA; LA is the logit-adjusted baseline, IETrans the released IETrans checkpoint scored on the same relations. The case-level interval is image-clustered. ΔAUC is the change in within-cell AUC, which no calibration or group-level shift can move (LA’s residue is float32 rounding); its interval is an image bootstrap.

	mR	ΔmR	group	case, 95% CI	ΔAUC , CI
TDE	.2476	+0.0972	+0.0841	+0.0130 [+0.0093, +0.0168]	+0.0026 [−0.0028, +0.0086]
LA	.2343	+0.0941	+0.0802	+0.0139 [+0.0101, +0.0176]	+0.0001 [−0.0000, +0.0001]
TDE–LA		+0.0031	+0.0039	−0.0008 [−0.0050, +0.0033]	+0.0026 [−0.0029, +0.0086]
IETrans	.3587	+0.2204	+0.2143	+0.0060 [−0.0031, +0.0151]	−0.0093 [−0.0181, +0.0004]

An operating-point control. We add one model that needs no training: the baseline’s logits minus $\tau \log \pi(y)$, the training predicate prior, with $\tau = 1$ [26]. Within a cell this adds the same constant to every log-odds $\log q(y) - \log q(z)$, so it changes no within-cell ranking: whatever case-level part it shows is existing discrimination spent at a new operating point, not new discrimination. It reaches $mR@50 = 0.2343$ at $R@50 = 0.6437$.

Mean recall, split. Pooled over identified relations, mean recall@50 rises by 0.0972 (0.1017 on the official per-image average). Of that, +0.0841 is group-level and +0.0130 case-level, interval [+0.0093, +0.0168] (Tab. 1): 87% of the gain survives shuffling every label among the relations that share its object pair. It lives in the body of the distribution. Splitting predicates by training frequency (9 head

above 10,000 instances, 26 body, 15 tail below 500), the body gains +0.1224, of which +0.1082 is group-level; the head loses 0.0255; the tail gains 0.0003. Of the 15 tail predicates, 1,480 test relations, the baseline recalls none and TDE recalls one, to, at 0.016: none of the ten points comes from the rarest predicates. Figure 3 shows the mechanism one predicate at a time. Recall of parked on rises from 0.000 to 0.886, almost all of it group-level: TDE now says parked on for (car, street), and so recalls most relations that carry it. Recall of on falls by 0.558 in the same way.

The case-level part is matched by the control. The control reproduces the whole picture: +0.0941 of mean recall, +0.0802 group-level and +0.0139 case-level. Against it, TDE’s case-level part is −0.0008, interval [−0.0050, +0.0033]. So a case-level part of a thresholded metric is not by itself evidence of new discrimination. When the boundary between on and parked on moves, the cases that cross it are those the baseline already scored as more parked on-like, and they are more often labelled so; the covariance is real, but bought with discrimination the baseline had. The direct test is the within-cell AUC, which no prior shift or recalibration can move: TDE’s change is +0.0026, interval [−0.0028, +0.0086].

The proper score agrees. With each model’s temperature fitted on held-out images ($T^* = 2.50$ baseline, 0.92 TDE), TDE’s quadratic-score change is −0.18264, of which −0.18258 is group-level; the case-level part is −0.00006, interval [−0.00120, +0.00109] (Fig. 2). Under the log score it is +0.04396 against a group-level −0.59157. Against the control the two scores disagree in sign (−0.0040 quadratic, +0.0154 log, both intervals excluding zero), and the control’s own case-level parts (+0.0039, +0.0285) show that the proper-score split, too, responds to the operating point

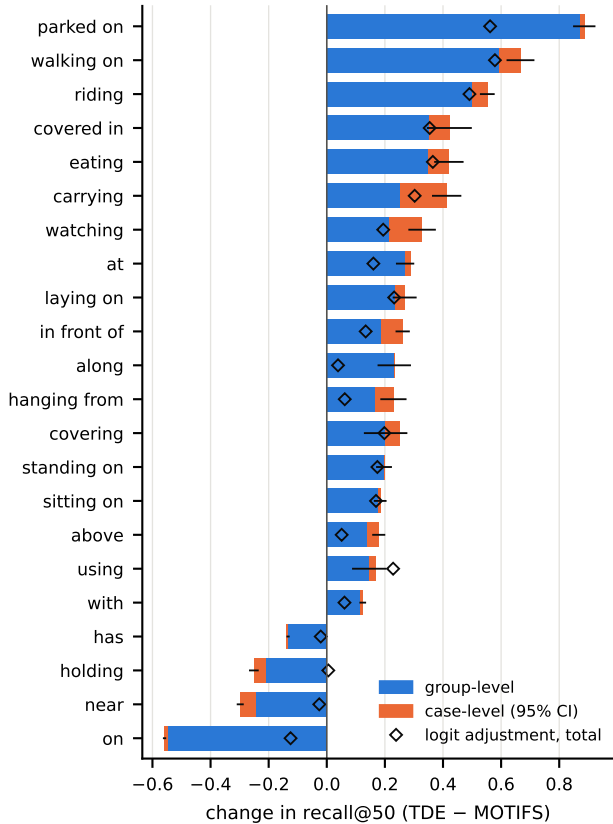


Figure 3. **Where TDE’s mean recall went**, for every predicate whose recall@50 moved by at least 0.12: group-level part (blue), case-level part (orange) with its image-clustered 95% interval, and the total under logit adjustment of the baseline (diamond), which changes no within-cell ranking.

(Appendix O). Unmatched outputs and a coarser grouping give different answers; all configurations are in Appendix O. Over 20 calibration halves TDE’s quadratic case-level part stays within $[-0.00131, +0.00059]$, never significantly positive; a temperature shared by both models makes it negative (Appendix Q.1). Merging synonymous predicates (32 classes) or collapsing them into three families leaves it null and IETrans’s negative (Appendix Q.2).

What that licenses. On this checkpoint, the ten points of mean recall are a change in which predicates the model favours within each object pair. Everything they did — the gain, its split, its tier profile — a one-line prior shift that ranks the relations of every cell exactly as the baseline did also does, at under a tenth of TDE’s cost in $R@50$ (0.0175 against 0.2023). Measured against that control, neither recall nor the threshold-free AUC finds any within-cell discrimination TDE gained at the benchmark’s label mix (Sec. 6), and the two proper scores split on its sign.

Why. TDE’s change from the baseline is three subtractions, and splitting along them answers why; the parts add up to TDE’s exactly (Appendix O.1). Removing the frequency prior, constant within every cell, moves mean recall by -0.0009 and the within-cell AUC not at all. Removing the visual term adds $+0.0164$, with a case-level part of $+0.0016$, interval $[-0.0006, +0.0037]$: within a cell, the context term already knew what the visual term did. Subtracting the averaged context supplies $+0.0817$ of the 0.0972. That term is, to 99.96% of its energy, one vector shared by every relation in every image, correlated $+0.75$ with the log training prior. In PredCls, TDE’s counterfactual is to first order a learned logit adjustment of the context branch, which is why the explicit one reproduces it.

A second method. IETrans [43] relabels training data, moving general predicates to specific ones within the same object pair, and adds the training log-prior at test time. We ran its released Neural-Motifs PredCls checkpoint once with the authors’ evaluation command and re-scored it the same way (Appendix O.2). It reaches $mR@50 = 0.3587$ (0.3576 on the authors’ test set, which drops duplicate relations): $+0.2204$ over the baseline, of which $+0.2143$ is group-level and $+0.0060$ case-level, interval $[-0.0031, +0.0151]$. Unlike TDE it reaches the tail ($+0.0889$), again group-level. Its case-level part does not improve and, on both proper scores, falls: matched case-level parts -0.0162 (quadratic) and -0.0544 (log), intervals excluding zero; the AUC change is -0.0093 , interval $[-0.0181, +0.0004]$, points the same way without excluding zero. The split separates the two methods: both gains are group-level, TDE’s with no case-level change, IETrans’s with a case-level loss. With no IETrans baseline released, this compares it with a separately trained model of its family.

6. The model every metric ranks last

The predictors of Sec. 4 never see an image: they read the class pair and, at most, box coordinates from the annotation. On the same reconstructed split we add one that reads pixels — subject and object regions cropped from the image and encoded by a frozen CLIP ViT-B/32 [29], concatenated with the class embeddings MLP-CLASS uses. CLIP is not fine-tuned, so any case-level gain reflects what pretraining had already recovered. All three compared models are retrained on one fixed subsample of 100,000 training relations, so differences isolate the features rather than the amount of data.

By every aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6156 against 0.6467 for the geometry model, and a quadratic score 0.04642 lower. An evaluator reading either number would discard it. The split disagrees. With each model’s confidence matched on held-out images, the CLIP model loses 0.04705 of group-level

score against the geometry model while *gaining* 0.00464 of case-level score (CI [+0.00311, +0.00618]); on raw outputs the figures are 0.05282 and 0.00640 (CI [0.00513, 0.00767], $p = .002$). Its whole deficit is group-level (Fig. 2). Against the class-only model, pixels buy 0.01663 of case-level score on raw outputs where box geometry buys 0.01023, with well-separated intervals.

A larger case-level part is not the same as better ranking, and here the two disagree. With each model’s prior matched within cells, the CLIP model still ranks the correct predicate first less often than the geometry model: top-1 accuracy 2.17 points lower, MRR 1.67 lower, recall@5 1.07 lower. So the CLIP model carries more case-level covariance without being the better ranker. The within-cell AUC, which no calibration or prior shift can move, does not settle it either way: 0.5408 against 0.5323, a difference of +0.0085 with interval [−0.0062, +0.0214], while both models clear the class-only model’s 0.5 by +0.0408 and +0.0323. Its case-level advantage is real covariance; that it is better recognition is not established.

Seeds. Over five training seeds (seed 0 reproduces the predictions above exactly; Appendix P), the CLIP model’s matched case-level advantage over geometry stays between +0.00413 and +0.00474, every interval excluding zero, its accuracy stays below geometry’s, and its AUC difference is positive in all five but excludes zero in one. Adding the CLIP features to geometry, rather than replacing it, doubles that advantage (+0.00807 to +0.00949). Zero-shot, with prompts and no training on Visual Genome, CLIP carries little of this: on the canonical relations of Sec. 5 its within-cell AUC is 0.5164 against MOTIFS’s 0.5705, and its case-level part over the frequency prior +0.00046 against +0.02943 (Appendix Q.4).

Under a label shift. A label shift should move group-level parts and only reweight case-level ones (Sec. 3). We test this by weighting the test relations so that every predicate of an object pair is equally frequent (Appendix Q.3). Across the eight comparisons of Sec. 5 and this section, seven group-level parts change sign, and the totals follow: the CLIP model now beats geometry by +0.02478. Five of the six case-level parts that excluded zero keep their sign and grow. The exceptions are TDE’s: against the baseline its null becomes +0.01430, interval [+0.01144, +0.01632], and against the control −0.00398 becomes +0.00723, while the AUC separates it from the baseline at neither mix. TDE gains discrimination between some predicate pairs and loses it between others, and the benchmark’s label mix weights the two to zero.

Beyond scene graphs. Nothing in Sec. 3 is specific to relations, and the supplementary applies the identical esti-

mator to long-tailed image classification on CIFAR-100-LT across seeds and imbalance ratios, with label-derived superclasses beside a global cell as the grouping, and to long-tailed text classification. On CIFAR-100-LT two standard re-weighting schedules have opposite compositions once confidence is matched, reversing what their raw scores suggest about both. The constraint on the grouping bites here: the fine class label cannot serve as a cell, because it leaves every cell label-homogeneous and forces the case-level part to zero by construction — an artefact that reads exactly like a null result.

7. Limitations and conclusion

Limitations. Four bound what the split can say. *It is conditional on the grouping.* The case-level part measures discrimination among relations sharing an object pair; two equally fine partitions of the same cases can disagree completely about whether a gain is case-level (Appendix K), so declaring ϕ is a modelling decision that the benchmark, not the submitting author, should make before a leaderboard opens. *It credits any within-cell signal,* including shortcuts nobody recorded, and our own bound on that exposure is not reassuring: for the controlled-predictor pair of Sec. 4, an unrecorded covariate correlated at 0.061 with both parts would account for the entire case-level gain (Appendix I). *It is not invariant to recalibration, to the operating point or to the label mix:* we read it confidence-matched and, for thresholded metrics, against an operating-point control; the within-cell AUC, invariant to the first two, gives up exact additivity; and a case-level null can be a cancellation that another label mix undoes (Sec. 6). *And it needs scores for every label:* most benchmarks archive only ranked top- k predictions, and recovering the rest took a rerun of the released checkpoint: that, not compute, blocks routine use.

Conclusion. A ten-point mean-recall gain is two different achievements summed together, and the field’s metrics report the sum. Scoring each prediction against another case’s label from the same group separates them exactly, with no fitted reference model and with image-clustered intervals. Applied to two released debiasing methods, the split finds mean-recall gains that are almost entirely group-level: TDE’s, whose case-level remainder a ranking-preserving logit adjustment matches and behind which no within-group discrimination was gained, and IETrans’s, with a case-level loss; applied to a model every aggregate metric ranks last, it finds a case-level gain that the aggregates conceal. We suggest that benchmarks whose labels lean on a feature every model sees report the split beside the headline metric: the declared grouping and its coverage, both signed parts with the case-level interval, the confidence-matched row, and, for a thresholded metric, the operating-point control. Code, results files and the scripts that regenerate every prediction are in the supplement.

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